Faculty of Computing and Information Technology
BMSE3153 Software Project Management
Assignment Report (202601)

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1. Project Background

1.1 Introduction

As the job market gets more competitive, the main issue the organization faces is attracting and securing top-tier employee candidates. The traditional recruitment methods are unable to address the demands of the modern workforce. The traditional recruitment methods usually involve time-consuming and repetitive manual processes such as email-based applications, spreadsheet tracking, and phone screenings. Businesses will struggle to manage high volumes of job postings, applications, and candidate screenings with these manual processes. Implementing the traditional methods may cause them to delay their hiring decisions, miss opportunities, and might even lose qualified candidates to competitors. These challenges are critical and need to be addressed. The need for a digital solution is higher than ever as we can automate and simplify the recruitment process.

The JobOnline Recruitment system that will be developed will be a web-based system that is developed to be able to streamline and optimise the hiring processes for organisations. The system aims to provide a centralised hub that connects job seekers directly with employers. This enables candidates to easily and effectively present themselves for the best-fit available opportunities. By having a modern web application with intuitive user interfaces, JobOnline is able to create a better experience for job seekers and employers. For employers, they will benefit from a centralized dashboard to manage job postings, allowing them to review all candidate applications and filter candidates based on their criteria. On the other hand, job seekers will be able to gain access to a user-friendly portal where they can search for opportunities that they might be interested in based on their skills. They are also able to create a professional profile, upload resumes, submit applications, and track their application statuses in real time.

The JobOnline system is guided by multiple high-level goals that allow us to measure its success which include automating repetitive and redundant tasks. These include application screening, interview scheduling, and candidate communication. Next, we plan to provide a platform that is intuitive, responsive, and accessible for both employers and job seekers. After that, we want to make a system that is scalable. One of the main problems with the traditional process was that it could not handle the sheer amount of volume of employers and job seekers, but with this modular system, we are able to handle this volume and more efficiently. Ultimately, JobOnline aims to be a transformative solution that addresses the problems that traditional systems have in today's competitive job market.

1.1.1 Project development team

Team Name: FusionTech Solutions

FusionTech Solutions is a software development team that specializes in the design and development of modern web-based applications. This team has 4 years of combined experience in software engineering, web development, database management, system analysis, and project management.

Our team is dedicated to creating and delivering innovative and reliable digital solutions that help tackle real-world business challenges. Our team of software engineering professionals and developers excels in different aspects of the software development lifecycle. This includes requirements analysis, system design, frontend and backend development, database administration, quality assurance, testing, and deployment. By merging technical knowledge with effective collaboration and communication, FusionTech Solutions is able to develop systems that are secure, scalable, and user-friendly.

For the JobOnline project, FusionTech Solutions is responsible for planning, designing, developing, and testing a web-based job recruitment platform that connects employers and job seekers through a centralized online system. The team focuses on implementing industry best practices in software engineering to ensure that the system delivers a seamless user experience while maintaining data integrity, security, and performance.

The development approach emphasizes maintainability, scalability, and usability, enabling the system to support growing numbers of employers, job postings, and job applicants. Through a structured project management process and a commitment to quality, FusionTech Solutions aims to deliver a recruitment platform that streamlines hiring activities, improves communication between stakeholders, and enhances overall recruitment efficiency.

FusionTech Solutions strives to provide a professional and dependable solution that meets stakeholder expectations and contributes to the digital transformation of recruitment processes.

1.1.2 Project client

The organization that will be sponsoring the project is JobOnline Sdn Bhd of Kuala Lumpur, Malaysia, which is a manpower consultancy firm operating on a small scale. This organization was established back in 2015 and was initially a small business focused on offering administrative staff in the region. Over the past decade, the organization has grown to become a regional organization with over 150 clients across Southeast Asia.

JobOnline is an innovative company working in the sphere of human resource technologies, whose goal is to assist the best candidates in finding the best companies. JobOnline is famous for personalized matching; however, because of rapid development, it becomes increasingly difficult to offer this service to its users. At the moment, the company has 25 people in its workforce.

Even though JobOnline has been successful in marketing their services, the internal processes at the company are still quite manual and they have led to a number of issues within the company. At the moment, the company uses an unorganized system of folders, e-mail attachments, and physical filing cabinets where thousands of candidate CVs are stored. There is no efficient way of accessing the information on candidates because of the current approach that the company uses. Also, traditional ways of communication such as telephone conversations and e-mails are used when talking to employers and job applicants, which leads to considerable delays in the recruiting process. Due to this fact, the company faces problems in hiring new employees and the recruiting process can take up to 30 days.

Additionally, keeping in mind their plan to expand to meet their next round of growth in which they hope to double their candidate pool by 2027, the management understands that their existing legacy systems stand in their way of competing with other larger fully-digitized recruitment organizations. In order to overcome these problems and maintain their efficiency, JobOnline is seeking out a robust and centralized system that works through the web to help with their recruitment process.

1.1.3 Nature of business and reasons

The JobOnline system is a system that operates within the human resource technology sector (HRTech). We have experienced the rapid growth of business processes being digitalised in recent years, and in this sector, the company's primary business model includes providing recruitment services and solutions to clients. These clients span across various industries, such as the information technology industry, the healthcare industry, the finance industry, the manufacturing industry, and the retail industry. In short, all industries that require candidates to become employees. JobOnline currently plans to serve approximately 150 client organisations and companies. We also aim to manage recruitment processes for an average of 500 job openings monthly for these companies and organisations.

Currently, the JobOnline company's business operation involves a 25-person team, which includes recruitment specialists who manually manage each job posting, application processing from all over the world, and candidate communication. This manual system has become unsustainable as the business is unable to handle the sheer amount of job postings and applications while simultaneously helping business owners communicate with candidates. It has become repetitive and increasingly time-consuming. The lack of a software system to automate these processes has resulted in inconsistent and redundant processes, data loss, and increasingly difficult tracking of key performance metrics. Clearly, there are multiple reasons why developing the JobOnline Recruitment is beneficial, including:

- **Significantly reduce the time-to-hire.**

By creating the JobOnline system, we are able to automate redundant and repetitive tasks to reduce wasted time and effort. These mainly include tasks such as application screening, interview scheduling, and candidate communication. The efficiency gained by the system will allow the specialists to focus on higher-value activities, which include candidate relationship building and strategic client consultation.

- **Improve data management and accessibility.**

We are able to achieve this by centralising all recruitment information into a database. We are able to implement a secure and searchable database that can be easily managed to reduce data loss. In addition to that, this system will not only eliminate the need for manual record tracking but also allow real-time access to candidate information across the organisation from anywhere and anytime. With this, we can also improve the data analytics and have better insights for the client's organisation.

- **Enhance the user experience.**

The JobOnline system can do this not only for the employers but also for the job seekers as well. For the employers, they will benefit from a streamlined process that allows them to post jobs and review applications. For job seekers, they will have better transparency and ease of use when using the JobOnline system. Having a system that encourages a positive user experience is essential for attracting and retaining employers and job seekers in the competitive recruitment market.

- **Support Security and Growth**

The system does this by providing a flexible platform. A platform that can accommodate increasing transaction demands and expanding services that are offered. The system will have modular components that can be enhanced and easily maintained. This allows the JobOnline system to be able to grow and meet market expansion requirements.

1.2 Project costs estimation

The total cost to develop the JobOnline: A Web-Based Job Recruitment System over a 3-month period has been estimated by breaking down all required resources into Manpower and Software. A 10% contingency reserve has been added to the base cost to cover unforeseen risks.

No	Category	July	August	September	Total (RM)
1	***Manpower***				
1.1	*Overall Project Manager*	*8,000*	*8,000*	*8,000*	*24,000*
1.2	*PM Assistant / Lead Developer (Team 1)*	*7,000*	*7,000*	*7,000*	*21,000*
1.3	*Business Analyst (Team 1)*	*5,500*	*5,500*	*5,500*	*16,500*
1.4	*UI/UX Lead (Team 1)*	*5,500*	*5,500*	*5,500*	*16,500*
1.5	*Developer (Team 1)*	*-*	*4,800*	*4,800*	*9,600*
1.6	*QA Lead (Team 1)*	*-*	*5,000*	*5,000*	*10,000*
1.7	*QA Engineer (Team 1)*	*-*	*4,200*	*4,200*	*8,400*
1.8	*PM Assistant (Team 2)*	*6,500*	*6,500*	*6,500*	*19,500*
1.9	*Business Analyst (Team 2)*	*5,000*	*5,000*	*5,000*	*15,000*
1.10	*UI/UX Lead (Team 2)*	*5,200*	*5,200*	*5,200*	*15,600*
1.11	*Lead Developer (Team 2)*	*-*	*7,000*	*7,000*	*14,000*
1.12	*Developer (Team 2)*	*-*	*4,500*	*4,500*	*9,000*
1.13	*QA Lead (Team 2)*	*-*	*4,800*	*4,800*	*9,600*
1.14	*QA Engineer (Team 2)*	*-*	*4,000*	*4,000*	*8,000*
1.15	*PM Assistant (Team 3)*	*6,500*	*6,500*	*6,500*	*19,500*
1.16	*Business Analyst (Team 3)*	*5,000*	*5,000*	*5,000*	*15,000*
1.17	*UI/UX Lead (Team 3)*	*5,200*	*5,200*	*5,200*	*15,600*
1.18	*Lead Developer (Team 3)*	*-*	*7,000*	*7,000*	*14,000*
1.19	*Developer 1 (Team 3)*	*-*	*4,500*	*4,500*	*9,000*
1.20	*Developer 2 (Team 3)*	*-*	*4,500*	*4,500*	*9,000*
1.21	*QA Lead / DevOps Engineer (Team 3)*	*-*	*6,000*	*6,000*	*12,000*

1.22	*QA Engineer (Team 3)*	*-*	*4,000*	*4,000*	*8,000*
	Subtotal - Manpower	*59,400*	*119,700*	*119,700*	*298,800*
2	***Software (Cloud Services)***				
2.1	*Amazon EC2 (t3.medium)*	*750*	*750*	*750*	*2,250*
2.2	*Amazon RDS (db.t3.small)*	*600*	*600*	*600*	*1,800*
2.3	*Amazon S3 (50GB + requests)*	*450*	*450*	*450*	*1,350*
2.4	*Data Transfer, Monitoring & Backup*	*600*	*600*	*600*	*1,800*
	Subtotal – Software	*2,400*	*2,400*	*2,400*	*7,200*
	Total Base Cost	*61,800*	*122,100*	*122,100*	*306,000*
	Contingency Reserve (10%)				*30,600*
	Grand Total				*336,600*

1.3 Project objective

The main objective of JobOnline is to build and complete the web based recruitment system by the 30th of September 2026 within a budget of RM 336,600. The aim of the system is to reduce the average hiring process from 30 days to less than 15 days through automated job postings, application management, candidate screening, interview scheduling, and reporting functionalities. The project objective will be described using the SMART criteria as follows:

- **Specific**

The objective is clear and easier to understand. The JobOnline web-based recruitment system is to be developed with these key functionalities which are job posting, application management, candidate screening, interview scheduling, and reporting as outlined in section 3.1. The system needs to be completed by the 30th of September 2026 within the budget of RM 336,600 with the business outcome of reducing the time process to increase efficiency by improving data management and accessibility through a centralised database, enhancing users experience for both employers and job seekers, and providing a secure and scalable platform to support future growth.

- **Measurable**

They are measures of effectiveness to display how successful the project is. The project is successful if:

1. It is completed by the 30th of September 2026 (within the project duration set).
2. The cost of development stays within the allocated budget of RM 336,600.
3. The average hiring process time is reduced to under 15 days.
4. The system achieves user satisfaction through the improved data accessibility and a user-friendly interface.

- **Achievable**

The objective is achievable based on the projects context. As mentioned in section 1.1.3, the current recruitment process takes about 30 days to hire. By creating this system to automate tasks like screening applications,

scheduling interviews, and communication with candidates through the JobOnline system, this can help reduce the hiring process to under 15 days. Additionally, the RM 336,600 budget is enough to support the development, testing, and deployment of the system as outlined in section 1.2. The centralised database will improve the data accessibility, and the design being modular will ensure scalability.

- **Relevant**

The objective will directly address the main business problem that is identified in section 1.1.2 and 1.1.3 where the current manual process is slow, inefficient, and unsustainable where the data is poorly managed across unorganised folder systems and email attachments. The user experience is also currently lacking for both the employers and job seekers and it cannot scale to support future growth. The JobOnline system will directly solve all these problems by providing clear benefits to all stakeholders and employers where they gain faster access to candidates suitable for the job, job seekers receiving a quicker response, and the client being able to scale their operations.

- **Time-constrained**

There is a clear deadline for when the objective must be achieved. The system has to be developed and deployed by 30th of September 2026. This timeframe gives the team a clear schedule, helps with planning and resource allocation, prevents any unnecessary changes to the project scopes, and allows the stakeholders to monitor the progress based on the defined timeline.

2. Project Team Structure

2.1 Team structure

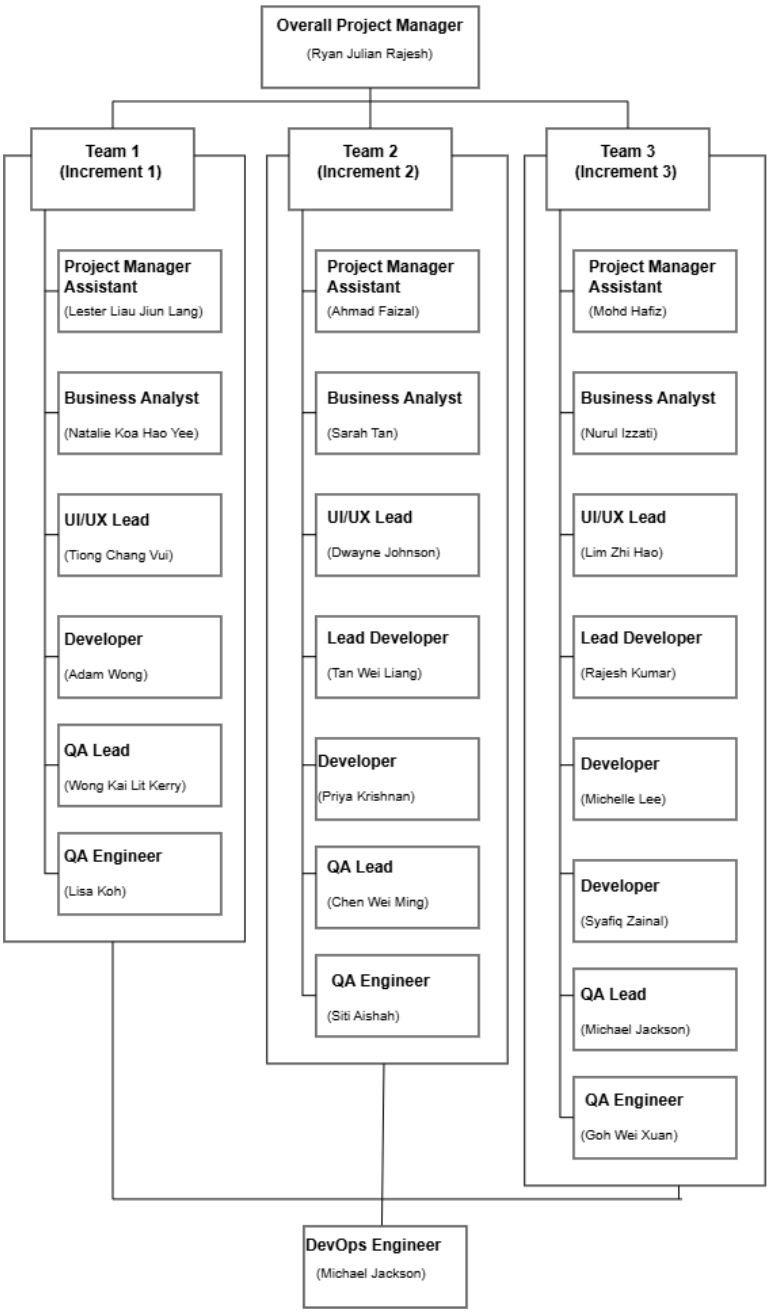


Figure 2.1: Organisational Chart

No	Name	Job Position	Task Allocated
1	Ryan Julian Rajesh	Overall Project Manager	● Leads the overall project ● Manages the timeline, budget, client communication, and risk ● Tracks milestones, allocates resources, and makes the final decisions
2	Lester Liau Jiun Lang	PM Assistant / Lead Developer (Team 1)	● Manages day-to-day coordination of Team 1 ● Designs system architecture and database schemas

			• Develops backend APIs and oversees code reviews
3	Natalie Koa Hao Yee	Business Analyst (Team 1)	• Gathers and documents requirements • Models business processes and develops use cases • Defines functional specifications and traces requirements
4	Tiong Chang Vui	UI/UX Lead (Team 1)	• Designs user interfaces and creates wireframes • Develops prototypes and conducts usability testing • Ensures responsive design and accessibility compliance
5	Adam Wong	Developer (Team 1)	• Builds full-stack features and implements functionality • Fixes bugs and documents code • Writes unit tests and optimises code
6	Wong Kai Lit Kerry	QA Lead (Team 1)	• Develops test strategies and creates test cases • Oversees manual and automated testing • Tracks defects, reports issues, and monitors quality metrics
7	Lisa Koh	QA Engineer (Team 1)	• Executes tests and performs regression testing • Conducts performance testing and security testing • Verifies bug fixes and documents test results
8	Ahmad Faizal	PM Assistant (Team 2)	• Manages day-to-day coordination of Team 2 • Tracks progress and reports to the Overall Project Manager • Facilitates communication within the team
9	Sarah Tan	Business Analyst (Team 2)	• Gathers and documents requirements • Models business processes and develops use cases • Defines functional specifications and traces requirements
10	Dwayne Johnson	UI/UX Lead (Team 2)	• Designs user interfaces and creates wireframes • Develops prototypes and conducts usability testing • Ensures responsive design and accessibility compliance
11	Tan Wei Liang	Lead Developer (Team 2)	• Designs system architecture and database schemas • Develops backend APIs and oversees code reviews • Makes technical decisions and optimises performance
12	Priya Krishnan	Developer (Team 2)	• Builds full-stack features and implements functionality • Fixes bugs and documents code • Writes unit tests and optimises code
13	Chen Wei Ming	QA Lead (Team 2)	• Develops test strategies and creates test cases • Oversees manual and automated testing • Tracks defects, reports issues, and monitors quality metrics
14	Siti Aishah	QA Engineer (Team 2)	• Executes tests and performs regression testing • Conducts performance testing and security testing • Verifies bug fixes and documents test results
15	Mohd Hafiz	PM Assistant (Team 3)	• Manages day-to-day coordination of Team 3 • Tracks progress and reports to Overall Project Manager • Facilitates communication within the team

16	Nurul Izzati	Business Analyst (Team 3)	● Gathers and documents requirements ● Models business processes and develops use cases ● Defines functional specifications and traces requirements
17	Lim Zhi Hao	UI/UX Lead (Team 3)	● Designs user interfaces and creates wireframes ● Develops prototypes and conducts usability testing ● Ensures responsive design and accessibility compliance
18	Rajesh Kumar	Lead Developer (Team 3)	● Designs system architecture and database schemas ● Develops backend APIs and oversees code reviews ● Makes technical decisions and optimises performance
19	Michelle Lee	Developer (Team 3)	● Builds full-stack features and implements functionality ● Fixes bugs and documents code ● Writes unit tests and optimises code
20	Syafiq Zainal	Developer (Team 3)	● Builds full-stack features and implements functionality ● Fixes bugs and documents code ● Writes unit tests and optimises code
21	Michael Jackson	QA Lead / DevOps Engineer	● Develops test strategies and creates test cases (Team 3) ● Sets up and maintains CI/CD pipelines ● Manages cloud infrastructure and configures servers
22	Goh Wei Xuan	QA Engineer (Team 3)	● Executes tests and performs regression testing ● Conducts performance testing and security testing ● Verifies bug fixes and documents test results

2.2 Stakeholders register

A list of all individuals involved in or is affected by the project. It helps the project team understand each stakeholders interests, level of influence, and communication needs throughout the project.

2.2.1 Project Client Stakeholders

No	Name	Title	Role	Expectation
1	Mr. Sam Liew Ee Xem	CEO	Project Sponsor	● The project to be delivered on time and within budget. ● The system to improve the overall recruitment process and provide a positive return on investment.
2	Mr Lim Jun Jie Jackson	Operations Director	Project Client	● The system is expected to simplify recruitment tasks and reduce manual workload. ● Regular progress update and timely delivery of the project's milestones.
3	Mr Ng Chan Fung Jason	HR Manager	End-User/ Product Owner	● The system to automate tasks that are repetitive such as application screening and interview scheduling. ● Improved hiring process efficiency.
4	Ms Raihan Jamal Wujdi	IT Manager	Technical Stakeholder	● System to be secure, scalable, and easy to maintain. ● Proper technical documentation and a smooth integration with the existing system.
5	Mr Liew Tong Ee	Senior Recruitment Specialist	End-User	● A user-friendly interface that simplifies job posting and candidate management. ● The system to reduce the time spent on administrative tasks.
6	Ms Quintinna Liew Syn Zhen	Recruitment Specialist	End-User	● The system provides quick access to the candidates information and easy communication with applicants. ● Reduced manual data entry.
7	Ms Elly Edward	Administrative Coordinator	End-User	● The system to generate accurate reports and track key recruitment metrics. ● A Simplified administrative processes.
8	Ms Bin Chi Lin Grace	External Consultant	Advisor	● The system to follow tech best practices. ● The system to be competitive with other digital recruitment platforms.

2.2.2 Project Development Team Stakeholders

No	Name	Role	Expectation
1	Ryan Julian Rajesh	Overall Project Manager	Project to be delivered on time, within budget, and meeting quality standards. Clear communication and timely decision making from members.
2	Lester Liau Jiun Lang	PM Assistant / Lead Developer (Team 1)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
3	Natalie Koa Hao Yee	Business Analyst (Team 1)	Clear and complete requirements from the client. Timely feedback and confirmation on requirement documentation.
4	Tiong Chang Vui	UI/UX Lead (Team 1)	Clear design requirements and feedback from users. Timely approval on wireframes and prototypes.
5	Adam Wong	Developer (Team 1)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
6	Wong Kai Lit Kerry	QA Lead (Team 1)	Completed features delivered on time for testing. Clear bug reporting and timely bug fixes from developers.
7	Lisa Koh	QA Engineer (Team 1)	Stable features delivered for testing. Clear test cases and proper testing environment.
8	Ahmad Faizal	PM Assistant (Team 2)	Clear task assignments and timely progress updates from team members. Effective coordination with Overall PM.
9	Sarah Tan	Business Analyst (Team 2)	Clear and complete requirements from the client. Timely feedback and confirmation on requirement documentation.
10	Dwayne Johnson	UI/UX Lead (Team 2)	Clear design requirements and feedback from users. Timely approval on wireframes and prototypes.
11	Tan Wei Liang	Lead Developer (Team 2)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
12	Priya Krishnan	Developer (Team 2)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
13	Chen Wei Ming	QA Lead (Team 2)	Completed features delivered on time for testing. Clear bug reporting and timely bug fixes from developers.
14	Siti Aishah	QA Engineer (Team 2)	Stable features delivered for testing. Clear test cases and proper testing environment.
15	Mohd Hafiz	PM Assistant (Team 3)	Clear task assignments and timely progress updates from team members. Effective coordination with Overall PM.
16	Nurul Izzati	Business Analyst (Team 3)	Clear and complete requirements from the client. Timely feedback and confirmation on requirement documentation.
17	Lim Zhi Hao	UI/UX Lead (Team 3)	Clear design requirements and feedback from users. Timely approval on wireframes and prototypes.

18	Rajesh Kumar	Lead Developer (Team 3)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
19	Michelle Lee	Developer (Team 3)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
20	Syafiq Zainal	Developer (Team 3)	Well-defined tasks and clear technical guidance. Proper development environment and tools to perform coding tasks.
21	Michael Jackson	QA Lead / DevOps Engineer (Team 3)	Stable code deployments and clear deployment schedules. Proper access and infrastructure to set up CI/CD pipelines.
22	Goh Wei Xuan	QA Engineer (Team 3)	Stable features delivered for testing. Clear test cases and proper testing environment.

2.3 Communication plan

Client Communication Plan

No	Name of The Communication (Method)	Purpose	Frequency	Audience	Owner
1.	Project Kickoff Meeting (Face-to-Face Meeting)	To formally initiate the project, align on objectives, define scope, introduce the core team, and establish the project governance framework.	Once (Project Initiation)	Project Sponsor, Operations Director, HR Manager, IT Manager, End-users	Ryan Julian Rajesh (Project Manager)
2.	Weekly Status Report (Email)	To provide a formal summary of completed work, visualize progress of the plan, highlight any critical risks or issues, and present the coming week's plan to the client/stakeholders	Weekly (Every Friday)	Project Sponsor, Operations Director, HR Manager, IT Manager, External Consultant	Ryan Julian Rajesh (Project Manager)
3.	Monthly Progress Review Meeting (Virtual Meeting - Google Meet / Zoom)	To evaluate overall project performance against the approved schedule and budget, and to realign strategies if needed	Monthly	Project Sponsor, Operations Director, HR Manager, IT Manager, External Consultant	Ryan Julian Rajesh (Project Manager)
4.	User Acceptance Testing (UAT) Session (Face-to-Face Meeting)	To gather feedback from end-users and ensure the system meets business requirements before deployment.	Upon completion of key milestones	HR Manager, End-Users (Recruitment Specialists, Admin Coordinators)	Business Analyst (Natalie Koa Hao Yee)
5.	Formal Documentation & Approval (Email)	To serve as the official channel for sharing critical documentation, including meeting minutes, change requests, formal approvals, and contractual deliverables.	As Needed	All Client Stakeholders	Ryan Julian Rajesh (Project Manager)

Development Team Communication Plan

No	Name of The Communication (Method)	Purpose	Frequency	Audience	Owner
1.	Project Kickoff Meeting (Face-to-Face Meeting)	To build team consensus on the project vision, understand high-level technical requirements, clarify roles and responsibilities, and establish the workflow and tooling protocols.	Once (Project Initiation)	Project Manager, Business Analyst, UI/UX Designer, Lead Developer, Developers, QA Team, DevOps Engineer	Ryan Julian Rajesh (Project Manager)
2.	Daily Check In (Virtual Meeting - Google Meet / Zoom)	To conduct daily stand-up updates, review task progress, and resolve blockers quickly.	Daily (15 minutes)	Project Manager, Developers, UI/UX Designers, QA Team, DevOps Engineer	Dwayne Johnson (Developer)
3.	Task Tracking & Monitoring (Project Management Tool - Trello)	To serve as the single source of truth for task tracking, real-time progress monitoring, assignment of responsibilities, and updating the status of issues and bugs.	Continuous	Lead Developer, Developers, QA Team, DevOps Engineer	Lester Liau Jiun Lang (Lead Developer)
4.	Team Communication (WhatsApp/Telegram Group Chat)	To have quick informal communication, provide immediate updates, and coordinate urgent matters.	As Needed	Development Team Members	Project Coordinator (Dwayne Johnson)
5.	Weekly Progress & Issue Review (Face-to-Face Meeting)	To conduct an in-depth review of project progress, discuss critical technical issues, review milestone achievements, and make key decisions on prioritization and strategy.	Weekly (Every Friday)	Project Manager, Business Analyst, UI/UX Designer, Lead Developer, QA Lead, DevOps Engineer	Project Manager (Ryan Julian Rajesh)
6.	Design Review & Approval (Face-to-Face Meeting)	To review wireframes, prototypes, and UI mockups, gather feedback, and secure approval before development begins.	As Needed (Per Sprint)	UI/UX Designer, Business Analyst, Lead Developer, Developers	Tiong Chang Vui (UI/UX Designer)
7.	Defect Triage & Bug Review (Virtual Meeting - Google Meet/Zoom)	To review reported bugs, prioritize fixes, assign ownership, and track defect resolution progress.	As Needed (Post-Testing)	QA Lead, QA Engineer, Lead Developer, Developers	Wong Kai Lit Kerry (QA Lead))
8.	Formal Documentation & Reporting (Email)	To share formal documentation, deliver official status reports, and maintain a record of technical decisions or escalations that require formal approval.	As Needed	All Development Team Members	Ryan Julian Rajesh (Project Manager)

3. Project Scope

3.1 Scope of the system

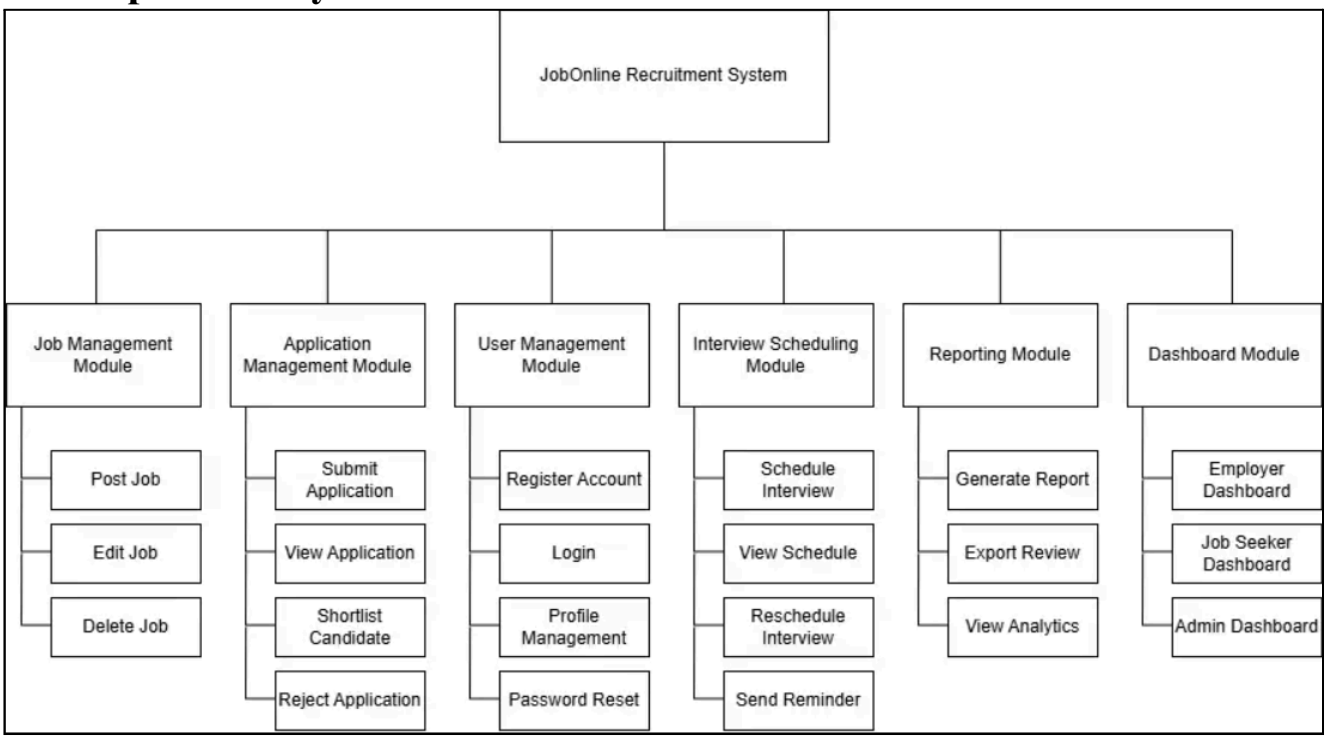


Figure 3.1: Scope of the JobOnline Recruitment System

No	Module	Function	Description
1.	Job Management	Post Job	Allow employers to create and publish job vacancies.
		Edit Job	Enables employers to update existing job postings.
		Delete Job	Allows employers to remove job postings that are no longer available.
2.	Application Management	Submit Application	Allows job seekers to apply for jobs by submitting their resume and cover letter.
		View Application	Enables employers to view all applications received for a specific job.
		Shortlist Candidate	Allows employers to shortlist qualified candidates for interview.
		Reject Application	Enables employers to reject unqualified applicants with automated notification.
3.	User Management	Register Account	Allow new users to create an account
		Login	Enables registered users to securely log in.

		Profile Management	Allows users to update their personal information and contact details.
		Password Reset	Provides a secure method for users to reset forgotten passwords.
4.	Interview Scheduling	Schedule Interview	Allows employers to schedule interviews with shortlisted candidates.
		View Schedule	Enables candidates to view their upcoming interview schedules.
		Reschedule Interview	Allows both employers and candidates to request and confirm rescheduling.
		Send Reminder	Automatically sends interview reminders via email.
5.	Reporting	Generate Report	Allows administrators to generate reports on recruitment metrics.
		Export Report	Enables users to export reports in PDF or Excel format.
		View Analytics	Provides visual analytics and charts on recruitment performance.
6.	Dashboard	Employer Dashboard	Provides employers with an overview of job postings and applications.
		Job Seeker Dashboard	Provides job seekers with an overview of their applications.
		Admin Dashboard	Provides administrators with system-wide analytics and user controls.

3.2 Exclusions

1. Hardware Procurement

The procurement, provision or installation of any physical hardware is not part of the development of the JobOnline recruitment system. The client (JobOnline Sdn Bhd) assumes full responsibility for their own servers, workstations, networking equipment and any other equipment they may need to host, access and maintain the system.

This project is just on Software Development, Testing and Deployment of a Web Based Recruitment System. As the system is designed to be hosted on the client's own infrastructure or on a cloud-based environment, the project team will not be involved in purchasing, setting up, or configuring any hardware components. This encompasses databases servers, application servers, load balancers, firewalls and end-user workstations, among others.

Before deployment, the client's IT team will be responsible for ensuring that the required hardware infrastructure is available to support the system's technical requirements. The client will be solely responsible for any hardware cost that arises (such as server maintenance, upgrade, repair and IT infrastructure setup) and will not be covered under the project budget of RM 140,525. Furthermore, the project team will not offer continuous hardware assistance and troubleshooting after software deployment.

2. Multi-Lingual Support

The JobOnline recruitment system will only be implemented in English. Multi-lingual support, including the translation of user interfaces, job postings, system messages, notifications, and reports into other languages such as Bahasa Malaysia, Chinese, or Tamil, is not included in the scope of this project.

The system will not support language switching functionality, and all content including job descriptions, candidate profiles, system communications, and administrative interfaces is expected to be submitted, stored, and displayed in English only. This decision is based on the current business context where the client, JobOnline Sdn Bhd, primarily operates in English-speaking markets and serves clients who conduct their recruitment processes in English.

Any future multi-lingual enhancements, localization efforts, or translation services would require a separate project phase, additional budget allocation, and an extended timeline beyond the current 3-month development period. If the client wishes to expand into non-English speaking markets in the future, this functionality can be considered as part of a future system upgrade or enhancement project. However, it is explicitly excluded from the current scope to ensure the project remains focused, on schedule, and within the approved budget of RM 140,525.

3.3 Software process model

3.3.1 Selected Model: Incremental Development Model (Parallel Increments)

Incremental Development Model with Parallel Development was selected as the preferred model for the implementation of the JobOnline project. In accordance with this development methodology, at the stage of planning and design, the completed and approved requirements specification and UI design plan will be developed. Afterwards, the software development will be divided into two parallel, independent increments developed in parallel by sub-teams.

3.3.2 Diagram: Incremental Development Model

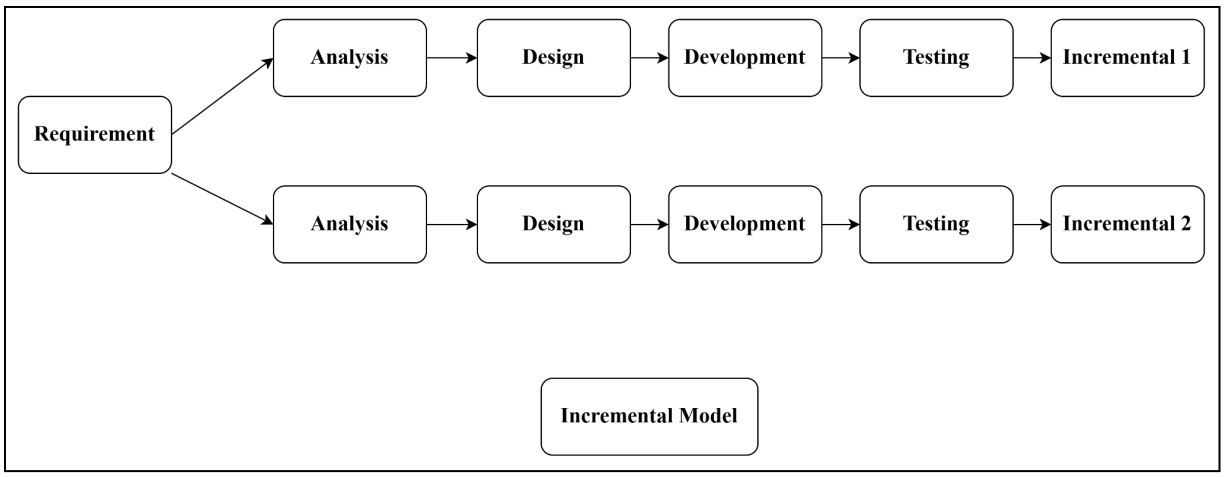


Figure 3.3.1

3.3.3 Description of the Model

In the Incremental Development Model, the system is subdivided into separate increments which go through the separate mini-lifecycle: requirements analysis, design, development and testing. Each increment is delivered as a completely functional product, on top of which next increment adds new functionality. The development of the system continues in such a manner until the whole system is developed.

Unlike the Waterfall Model, in the Incremental Development Model testing and integration take place through all stages of development rather than only at the end. Also, the Incremental Development differs from the Agile Development Model by fixing of the requirements for each increment before the start of development. As a result, more predictability is provided, but the client receives working software in the early stages of the project.

At the early stages of project implementation, the increment contains all the mandatory features of the product, and further increments add additional, supporting or less important features. After the delivery of each increment, the client is able to review it and give his feedback for the next increment.

3.3.4 Advantages of the Incremental Model

Advantage 1: Early and Regular Delivery of Working Software
Each increment is developed independently and is a fully tested software which is already ready for use by the client. It gives the client a possibility to gain his confidence in the correctness of the development and allows training users on the ready software.

Advantage 2: Easier Risk Management
The project is divided into separate, well-controlled increments, thus any possible problems in one increment do not affect the entire project. Also, the risk of appearing serious bugs at the final stage of development is significantly reduced due to the fact that at each increment testing takes place separately for each small part of the software.

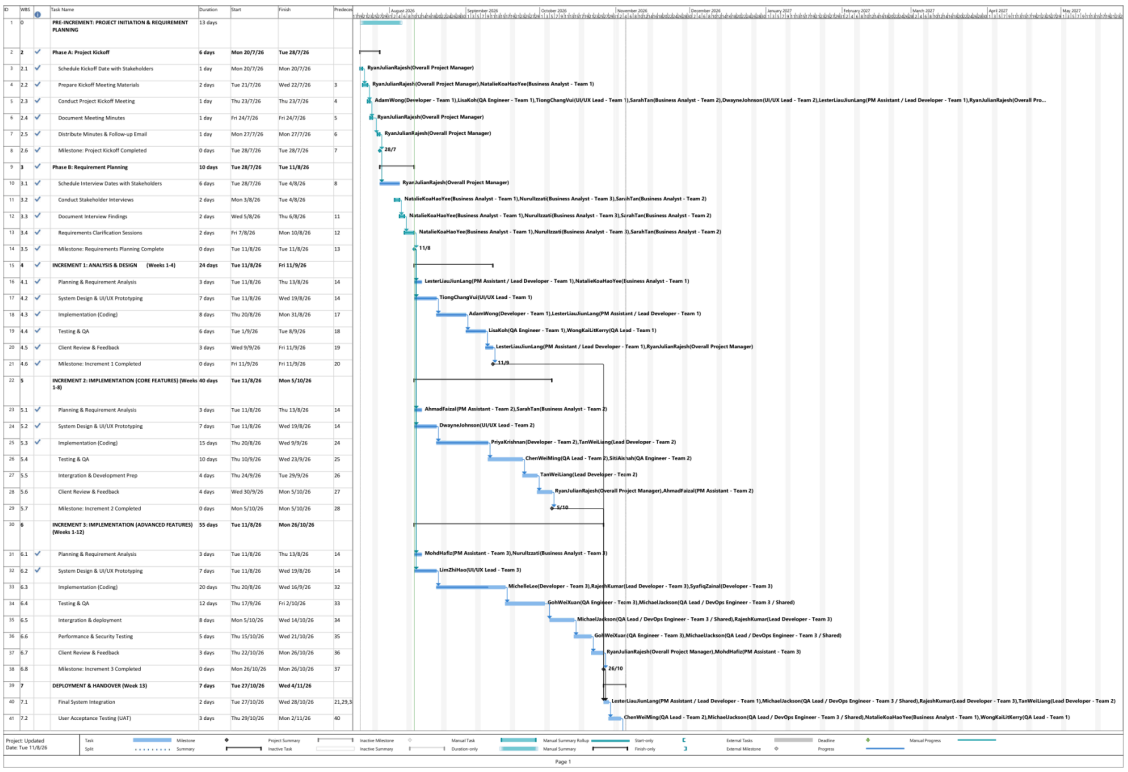
3.3.5 Disadvantages of the Incremental Model

Disadvantage 1: Rigid Requirement Freeze and Integration Challenges in Increments
As soon as the requirements for an increment have been defined and development has begun, there is a requirement freeze, meaning that further changes in the requirements can happen only in the next increment. Also, if the client needs some changes after the completion of the first increment, he or she will have to wait until the next increment is completed. Besides that, as increments are being developed in parallel and share frozen requirements, integration of these sub-teams' work can become problematic due to possible integration bottlenecks.

Disadvantage 2: Need to Proper Planning and Design of the System Are Required
As the entire process consists of several separate parts that should be integrated into one project later, the design of the whole system should be elaborated properly at the beginning stage. Otherwise, the integration of some increments might be impossible.

4. Project Schedule

4.1 Gantt Chart





4.2 Assumptions

The following assumptions were made during the planning of the JobOnline project. These assumptions are used to form the basis for the project schedule, budget, and resource allocation. If any of these assumptions proves to be incorrect, the project plan may need to be adjusted accordingly.

1. **Stakeholder Availability**

All key stakeholders from the client side, including the project sponsor, operations director, HR manager, IT manager, recruitment specialists, and administrative coordinators, will be available throughout the project's duration as they are to provide timely feedback, attend scheduled meetings, and approve deliverables as planned. This includes availability during the requirements gathering phase which is during week 1 to 4, where their input is important and crucial for defining the systems functionalities, and during the User Acceptance Testing (UAT) phase where the end users validation is required before deployment. Any delays in stakeholder availability especially during these critical phases, could impact the project's timeline and may require schedule adjustments.

2. **Working Days Exclude Weekends and Public Holidays**

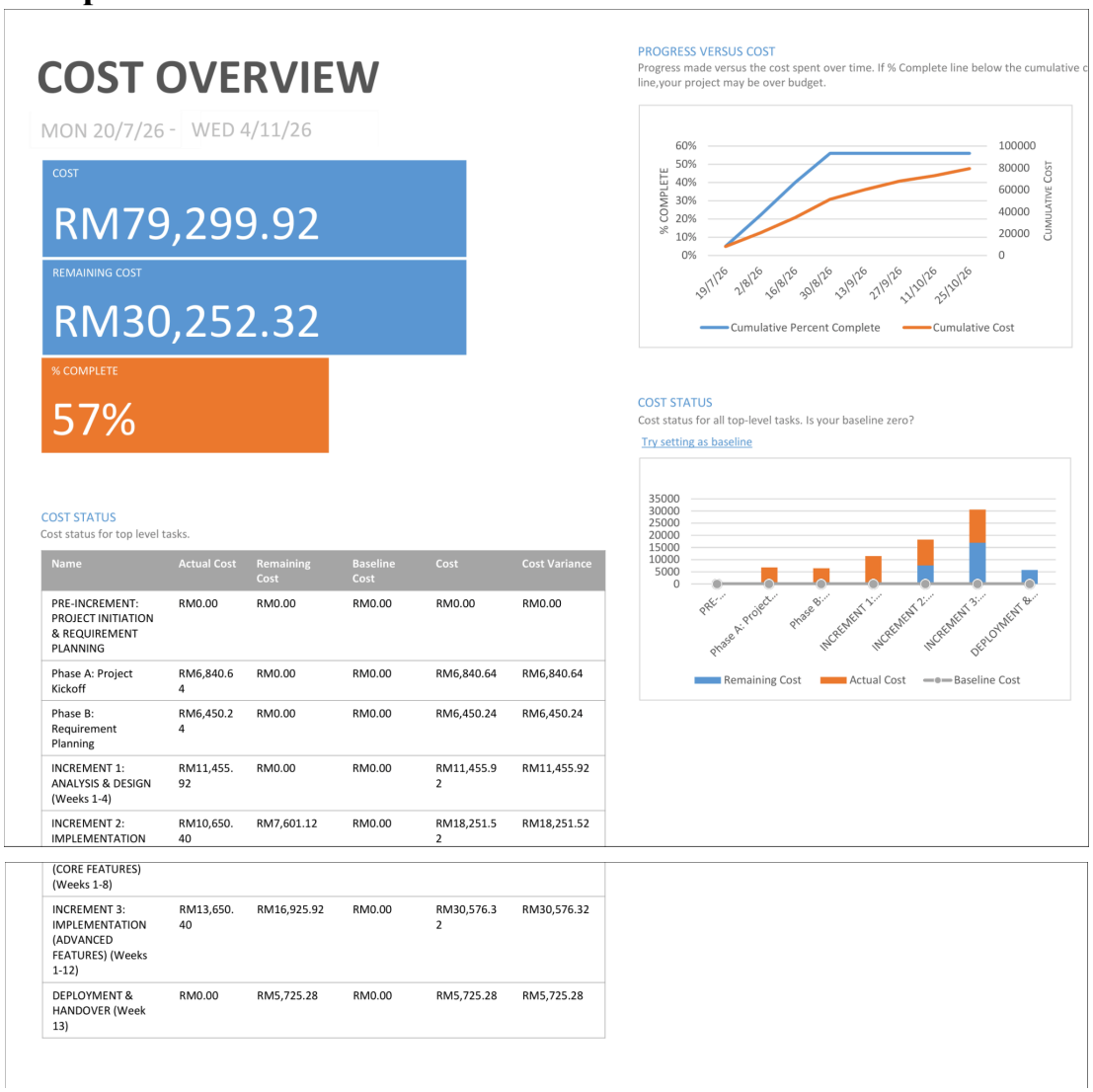
This project assumes a standard work schedule from Monday to Friday only. Saturdays, Sundays, and public holidays will not be considered as a working days and will be excluded from the projects timeline. The three-month project duration which is from July to September 2026, is calculated based on working days only. Any public holidays that fall within the project period, such as Malaysia's National Day on the 31st of August and Malaysia Day on the 16th of September, will not be counted as working days. The project team will not be expected to work on these days, and the project schedule in the Gantt chart reflects this assumption.

Conclusion

These assumptions have been documented as it is to provide transparency to the stakeholders and to establish a baseline for project planning. If any of these assumptions are found to be incorrect, the project manager will reassess the plan thoroughly and implement the appropriate corrective actions.

4.3 Reports

4.3.1 Cost Report

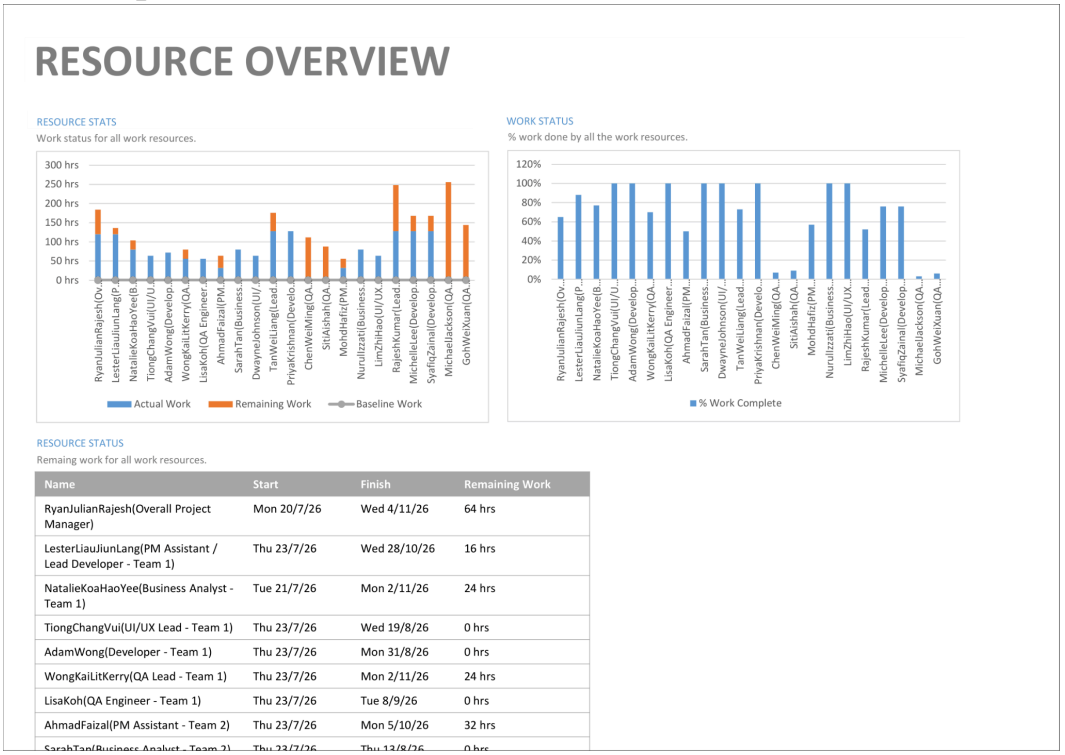


Name	Actual Cost	Remaining Cost	Baseline Cost	Cost	Cost Variance
PRE-INCREMENT: PROJECT INITIATION & REQUIREMENT PLANNING	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00
Phase A: Project Kickoff	RM6,840.64	RM0.00	RM0.00	RM6,840.64	RM6,840.64
Phase B: Requirement Planning	RM6,450.24	RM0.00	RM0.00	RM6,450.24	RM6,450.24
INCREMENT 1: ANALYSIS & DESIGN (Weeks 1-4)	RM11,455.92	RM0.00	RM0.00	RM11,455.92	RM11,455.92
INCREMENT 2: IMPLEMENTATION (CORE FEATURES) (Weeks 1-8)	RM10,650.40	RM7,601.12	RM0.00	RM18,251.52	RM18,251.52
INCREMENT 3: IMPLEMENTATION (ADVANCED FEATURES) (Weeks 1-12)	RM13,650.40	RM16,925.92	RM0.00	RM30,576.32	RM30,576.32
DEPLOYMENT & HANDOVER (Week 13)	RM0.00	RM5,725.28	RM0.00	RM5,725.28	RM5,725.28

The Cost Overview report shows the project's financial health and progress for the period of the project, which is from July 20, 2026, to November 4, 2026. This report is structured into two sections: a high-level dashboard summary and a granular, task-by-task financial breakdown.

The top half represents the project's three core Kep Performance Indicators. These are the Total Cost, which is RM79,299.92, the Remaining Cost, which is RM30,252.32, and the overall % Complete, which is 57%. The visual representations are highly useful as it helps stakeholders immediately assess the budget in a simple yet effective way. This is because we can clearly see that if the completion line falls below the cost curve, it is a signal that the budget is potentially overrun, which can give us time to take action and prevent the overrun of our budget. The bottom half of the report consists of the cost status table. It allows us to more easily track the actual cost, remaining cost, Baseline cost, and cost variance for each working package.

4.3.1 Resource Report



Name	Start	Finish	Remaining Work
RyanJulianRajesh(Overall Project Manager)	Mon 20/7/26	Wed 4/11/26	64 hrs
LesterLiauJiunLang(PM Assistant / Lead Developer - Team 1)	Thu 23/7/26	Wed 28/10/26	16 hrs
NatalieKoaHaoYee(Business Analyst - Team 1)	Tue 21/7/26	Mon 2/11/26	24 hrs
TiongChangVui(UI/UX Lead - Team 1)	Thu 23/7/26	Wed 19/8/26	0 hrs
AdamWong(Developer - Team 1)	Thu 23/7/26	Mon 31/8/26	0 hrs
WongKaiLitKerry(QA Lead - Team 1)	Thu 23/7/26	Mon 2/11/26	24 hrs
LisaKoh(QA Engineer - Team 1)	Thu 23/7/26	Tue 8/9/26	0 hrs
AhmadFaizal(PM Assistant - Team 2)	Thu 23/7/26	Mon 5/10/26	32 hrs
DwayneJohnson(UI/UX Lead - Team 2)	Thu 23/7/26	Wed 19/8/26	0 hrs
TanWeiLiang(Lead Developer - Team 2)	Thu 23/7/26	Wed 28/10/26	48 hrs
PriyaKrishnan(Developer - Team 2)	Thu 23/7/26	Wed 9/9/26	0 hrs
ChenWeiMing(QA Lead - Team 2)	Thu 23/7/26	Mon 2/11/26	104 hrs
SitiAishah(QA Engineer - Team 2)	Thu 23/7/26	Wed 23/9/26	80 hrs
MohdHafiz(PM Assistant - Team 3)	Thu 23/7/26	Mon 26/10/26	24 hrs
NurulIzzati(Business Analyst - Team 3)	Thu 23/7/26	Thu 13/8/26	0 hrs
LimZhiHao(UI/UX Lead - Team 3)	Thu 23/7/26	Wed 19/8/26	0 hrs
RajeshKumar(Lead Developer - Team 3)	Thu 23/7/26	Wed 28/10/26	120 hrs
MichelleLee(Developer - Team 3)	Thu 23/7/26	Wed 16/9/26	40 hrs
SyafiqZainal(Developer - Team 3)	Thu 23/7/26	Wed 16/9/26	40 hrs
MichaelJackson(QA Lead / DevOps Engineer - Team 3 / Shared)	Thu 23/7/26	Tue 3/11/26	248 hrs
GohWeiXuan(QA Engineer - Team 3)	Thu 23/7/26	Wed 21/10/26	136 hrs

The Resource Overview report helps provide a team-focused analysis of the project's workflow and its resources. It is divided into three sections which are the stacked bar chart that illustrates the resource statistics (which are Actual vs. Remaining vs. Baseline hours), a percentage bar chart that tracks work completion, and a detailed table listing the individual hours and schedules for every assigned team member. In conclusion, having these visual charts with granular table data, the management team can easily optimise resource utilisation and validate if the actual workload is deviating from the main baseline plan. With this, we can confidently estimate if we can get the project done on time.

5. Quality Plan

5.1 Quality Attribute

For the JobOnline recruitment system, the key software quality attribute is Performance Efficiency (Time Taken) , as defined by the ISO/IEC 25010 standard. Performance Efficiency refers to the response time and processing speed of the system when users perform tasks such as searching for jobs. This software quality attribute is critical for JobOnline because the system must process large volumes of job postings, applications, and user interactions simultaneously without compromising response times or user experience.

The JobOnline system is expected to serve approximately 150 client organisations and manage an average of 500 job openings monthly. With thousands of candidate CVs stored in the system and multiple users (employers, job seekers, and administrators) accessing the platform concurrently, performance efficiency is essential to ensure that users can complete tasks quickly and efficiently. Delays in system response would directly contradict the project's objective of reducing the average hiring process time from 30 days to less than 15 days.

The system's performance will be measured by the time taken (in seconds) to search for jobs. Load testing will be conducted using Apache JMeter with 50 concurrent users, and the average response time will be calculated using the formula: Average Response = Total Response Time / Number of Users. The maximum acceptable response time is 3.0 seconds (any average exceeding this triggers build rejection), with a target range of 0.5 to 2.0 seconds to ensure a positive user experience. The first performance test has not yet been conducted and is scheduled for Week 7 during Increment 2 of the development cycle.

Table 5.1: Software Quality Specification for Performance Efficiency

Performance Efficiency (Time Taken)	How quickly the system responds to user actions and completes tasks.
Scale	TIME taken (in seconds) for the system to: ● Search for jobs
Test	1. Execute automated load testing using Apache JMeter simulating **50 concurrent users** performing job searches. 2. Record the response time for each user's job search operation. 3. Calculate the average response time by using this formula: **Average Response = Total Response Time / Number of Users** (Total time taken by all 50 users divided by 50).
Max Acceptable	**3.0 seconds** for any operation. (Any average response time exceeding **3.0 seconds** triggers build rejection).
Target Range	**0.5 – 2 seconds** for all operations. This is to ensure that users (employers and job seekers) will adopt the JobOnline system and have a positive user experience.
Now	Not yet measured (system is still under development). The first performance test will be conducted in Week 7 during Increment 2 of the development cycle.

6. Risk Table

6.1 Risk

Risk Information Sheet				
Risk ID: R-01	**Date: 1 July 2026**	**Probability: 35%**	**Impact: High**	**Risk Category: TE**

Risk Description:
As a result of developing Increments A (Employer-centric features) and Increment B (job seeker & admin features) separately using the Parallel Incremental approach, there is a chance that the database schemas, API contracts, or authorization flows do not match perfectly. In such case, integration problems might occur during the merging process in Weeks 12-13 leading to delays after 30 September 2026.

Risk Mitigation:
1. Develop a single system architecture and database schema before beginning the development process. The Lead Developer (Team 1) shall develop a single technical design document which both teams have to follow. It will guarantee that all modules utilize the same interface and data structure from Day 1.

2. Assign the Lead Developer (Team 1) as overall technical lead of the project in order to enforce coding standard and API contracts. So he or she will be responsible for reviewing code of both increments and approving all the API endpoints for each week.

3. Can conduct cross-team design reviews at the beginning of each increment. Each sub-teams should discuss their features before the coding begins and will identify possible overlapping or conflicting components.

4. Implement a common Continuous Integration (CI) / Continuous Delivery/Deployment (CD) pipeline with integration testing automation starting from Week 5. The DevOps Engineer will configure the pipeline in such a way that integration testing will be conducted each time if new code is delivered.

5. Freeze the design of API and databases after the planning phase is finished. Should prevent changes to the interface will be allowed starting from Week 5 without the approval of the Overall Project Manager.

Risk Monitoring:
1. Weekly cross-team technical alignment meetings. Such meetings will allow both teams to share technical details, progress and decisions and identify any potential integration problem before it becomes a blocker.

2. Monitoring the results of integration testing after each build cycle. The QA Lead (Team 1) will be responsible for monitoring Continuous Integration (CI) / Continuous Delivery/Deployment (CD) pipeline output and notifying the team about any failed integration test.

3. Monitoring integration issues logged in Trello an online board with severity classification. Each issue identified will be logged, classified and reviewed on a weekly basis.

4. Checking API contract compliance via automated validation tools. The API contract validation will be done using tools such as Swagger/OpenAPI.

5. Reporting recurrent integration conflicts to the Overall Project Manager. If the same integration problem occurs more than twice it should be escalated.

Risk Management:
1. Set aside contingency reserve (RM 30,600) to deal with increased work of integration. This money can be spent on hiring more developers, paying for overtime, and using contractors in case of necessary integration rework.

2. Allocate only Week 12 to the integration process, and after the eleventh week, do not add any new features. This approach requires both teams to stop adding new features and focus exclusively on merging increments to prevent last-minute integration conflicts.

3. In case of failed integration, reduce the scope of non-critical features in Increment B (such as advanced reporting) and postpone their implementation to Phase 2. Reducing the scope of non-critical features will allow the team to address integration issues without postponing the system implementation.

4. Temporarily reassign the DevOps Engineer to support the Lead Developer with integration work. The DevOps Engineer has skills related to deployment and environment setup which makes him suitable for solving infrastructure-related integration issues.

5. In case the integration causes delays longer than 3 days, conduct an emergency meeting with the Project Sponsor to discuss changing the date of delivery or the scope. This approach involves the client in decision-making to find the solution to the problem and to preserve trust.

The mitigation strategies focus on preventing integration issues by setting technical standards, design agreement, and cross-team coordination. The monitoring strategies help to detect potential issues in time to address them while they are inexpensive to solve. Finally, the management strategies offer a backup option: in case of the occurrence of risks despite all preventive measures, there are buffers in terms of budget, schedule, and scope of work which can absorb negative impacts of the risk on the project.

7. Ethics and Professional conduct in project management

7.1 Engagement strategies with key stakeholders

Strategy 1: Engagement with the Project Sponsor / CEO (Mr. Sam Liew Ee Xem)

Strategy Name: Strategic Alignment and High-Level Progress Reporting

The Project Sponsor (CEO) is the most influential stakeholder and primary decision-maker for the project, responsible for budget approval and ensuring the project delivers business value on time and within budget. This engagement strategy focuses on providing concise, high-level progress updates and formal approval mechanisms without unnecessary technical details. This allows the CEO to maintain strategic oversight while the project team manages day-to-day execution.

Supporting Communication Methods:

- Project Kickoff Meeting (Face-to-Face Meeting)

How it supports the strategy: This meeting serves as the foundation for establishing a strong relationship with the CEO. It allows the Project Manager (Ryan Julian Rajesh) to align on project objectives formally, present the project vision, and secure the CEO's commitment. The face-to-face nature of this meeting demonstrates professionalism and respect for the CEO's authority while providing an opportunity to clarify expectations and address any concerns upfront.

- Weekly Status Report (Email)

How it supports the strategy: The weekly status report provides the CEO with a regular, predictable update on the project's health. The report summarises completed work, highlights any critical risks or issues, and presents the upcoming week's plan. This prevents the CEO from being caught off guard by unexpected problems and demonstrates proactive management.

- Monthly Progress Review Meeting (Virtual Meeting - Google Meet / Zoom)

How it supports the strategy: The monthly review meeting offers a deeper dive into project performance against the approved schedule and budget. It allows the CEO to evaluate whether the project is on track to deliver the expected return on investment (ROI) and provides a forum for strategic decision-making if course corrections are needed. This ensures the CEO remains confident in the project's direction.

- Formal Documentation & Approval (Email)

How it supports the strategy: The CEO must formally approve critical project decisions, such as scope changes or budget adjustments. Using formal email documentation creates a clear audit trail and ensures that all decisions are properly recorded. This protects both the project team and the CEO by providing transparency and accountability.

Strategy 2: Engagement with End-Users (HR Manager & Recruitment Specialists)

Strategy Name: User-Centric Validation and Continuous Feedback

The HR Manager (Mr. Ng Chan Fung Jason) and Recruitment Specialists (Mr. Liew Tong Ee, Ms. Quintinna Liew Syn Zhen, and Ms. Elly Edward) are the primary end-users of the JobOnline system. Their satisfaction is critical to the project's success as they will use the system daily. Therefore, they will be involved early and continuously to ensure the system meets their practical needs, is user-friendly, and reduces manual workload. This approach builds ownership, encourages acceptance, and reduces resistance and post-launch dissatisfaction.

Supporting Communication Methods:

- Project Kickoff Meeting (Face-to-Face Meeting)

How it supports the strategy: The kickoff meeting introduces end-users to the project team and establishes a collaborative tone. It helps end-users understand the project's purpose and how the new system will improve their daily work. This early involvement fosters a sense of ownership and reduces potential resistance to change.

- User Acceptance Testing (UAT) Session (Face-to-Face Meeting)

How it supports the strategy: UAT sessions are the most critical engagement point for end-users. By involving HR Manager and Recruitment Specialists in testing the system before deployment, they can provide direct feedback on usability, functionality, and whether the system meets their daily operational needs. This ensures that the final product is practical and aligned with their expectations, reducing the risk of dissatisfaction after launch.

- Weekly Status Report (Email)

How it supports the strategy: While end-users may not need the same level of detail as the CEO, the weekly status report keeps them informed about which features are being developed and when they can expect to use them. This transparency builds trust and helps end-users prepare for the transition to the new system.

- Monthly Progress Review Meeting (Virtual Meeting - Google Meet / Zoom)

How it supports the strategy: Inviting end-user representatives to these meetings gives them a voice in the project's direction. They can share feedback on how features are developing and raise any concerns about usability or missing requirements. This collaborative approach ensures the system evolves to meet their practical needs.

7.2 Reasons to disclose the disadvantages of the chosen model

The JobOnline project has adopted the Incremental Development Model with Parallel Increments in Section 3.3 to deliver the system within a tight three month schedule. While this model offers advantages such as faster delivery and better resource utilisation, but it has disadvantages that must be openly communicated to the stakeholders.

Managing Stakeholder Expectations
Stakeholders has the expectation that the system is to be completed by the 30th of September 2026 within budget. However, the Incremental Model has increased integration risk where Increment A and Increment B are developed independently and may not align perfectly during integration phase in Weeks 12 to 13. If not disclosed correctly, stakeholders will be unprepared by integration issues when it arises which then leads to frustration and loss of trust. Disclosure prepares them for any potential challenges and reduces disappointment when issues do arise.

Disclosing Requirement Rigidity
The model also has requirement rigidity where once an increment starts, requirements are frozened. If the client requests changes in the middle of the increment they must wait until the next increment. Without disclosure, stakeholders may see the team as inflexible and uncooperative. With disclosure, they understand that this limitation is a trade-off for faster delivery. By being honest it can aid in building trust and strengthens the client-vendor relationship, making stakeholders more cooperative during challenging phases like UAT and deployment.

Encouraging Informed Decision-Making
The model requires careful early planning and architecture must be well defined from the start. If stakeholders are unaware, they may rush through approvals, and assume changes can be made later. However, changes become costly once an increment is underway. Disclosure encourages that the stakeholders to invest adequate time in reviewing the requirements and mockups during the planning phase which is during Week 1 to 4. It also enables the IT Manager and External Consultant to raise any technical concerns early and contribute to the mitigation strategies.

Enabling Collaborative Risk Management
When stakeholders are aware of the model's disadvantages, they can cooperate with the project teams to address them. For example, knowing about integration risks, they can allocate more time for integration testing. Knowing about requirement rigidity, they can prioritise features carefully before each increment starts. This collaborative approach reduces the likelihood of any surprises and ensures smoother execution of the project.

7.3 Corrective actions (lester)

Assuming the JobOnline project is significantly behind schedule, the project manager must take decisive action to bring the project back on track. The following two corrective actions are proposed, considering the project's 3-month timeline and the 10-person team structure.

Corrective Action 1: Reassigning Tasks and Encouraging the Team

a) Reassigning Tasks: To remove bottlenecks and optimize the use of available skills, the Project Manager will temporarily reassign non-critical tasks. For example, the Lead Developer (Lester Liau Jiun Lang) will be relieved of administrative duties such as code review coordination and updating the task tracking board (Trello), which will be delegated to the Business Analyst (Natalie Koa Hao Yee). This will free up Lester's time to focus on resolving complex backend integration issues. Additionally, the DevOps Engineer (Michael Jackson) will temporarily assist the QA team with setting up automated test environments to accelerate the testing phase.

b) Encouraging the Team: To boost morale and motivation during this challenging period, the Project Manager will introduce small incentives and recognition. This will include issuing public acknowledgements during daily stand-up meetings for team members who complete their tasks ahead of schedule, providing vouchers for meals for the team once a week, and organizing a small celebration (e.g., team lunch) upon the successful completion of each major milestone (e.g., completion of both parallel increments).

Advantage:

1. Improved Efficiency and Focus: Reassigning administrative and supportive tasks to appropriate team members ensures that the lead developer's valuable time is spent on high-impact technical work. The incentives and encouragement will boost team morale, leading to increased focus, better collaboration, and a more positive work environment, which can directly translate to higher productivity and faster progress.

Disadvantage:

1. Risk of Reassignment Overload and Role Confusion: Delegating tasks to team members who are not initially responsible may increase their workload and lead to confusion about roles and responsibilities. For example, if the Business Analyst is already busy with requirement clarification, taking on additional administrative tasks may cause her work to suffer. This requires careful monitoring to ensure the reassignment does not simply create new bottlenecks elsewhere.

Corrective Action 2: Negotiating Changes in the Specification

The Project Manager (Ryan Julian Rajesh) and Business Analyst (Natalie Koa Hao Yee) will formally engage the client stakeholders, particularly the Project Sponsor (Mr. Sam Liew Ee Xem) and HR Manager (Mr. Ng Chan Fung Jason), to negotiate a temporary reduction in the project scope. This involves deferring specific low-priority features, such as the advanced visual analytics charts in the Reporting module or the automated interview reminder feature, to a post-launch Phase 2 update. The team will focus development efforts solely on the essential core functionalities: job posting, application management, user registration/login, and basic interview scheduling. A formal change request will be documented, and the project schedule will be re-baselined based on the agreed reduced scope, ensuring the system can be successfully deployed by the 30th of September deadline.

Advantage:

1. Realistic Delivery and Reduced Pressure: This action immediately aligns the project's deliverables with its available time and resources. By focusing on the "Must-Have" features, the development team can work without excessive pressure, maintaining higher code quality and reducing the risk of introducing critical bugs. This increases the likelihood of delivering a stable, functional system on time.

Disadvantage:

1. Potential Client Dissatisfaction and Perceived Loss of Value: The client may perceive this as a failure to deliver the full system they envisioned. Despite clear communication, the Recruitment Specialists may be disappointed that features they were looking forward to (e.g., advanced reporting) are not available at launch. This could strain the client-vendor relationship and negatively impact the project's perceived success.

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